

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Simple JS Calculator</title>

  <style>

    /* Simple styling to make it look clean */

    body {

      font-family: sans-serif;

      display: flex;

      justify-content: center;

      align-items: center;

      height: 100vh;

      background-color: #f4f4f9;

      margin: 0;

    }

    .calculator-card {

      background: white;

      padding: 2rem;

      border-radius: 12px;

      box-shadow: 0 10px 25px rgba(0,0,0,0.1);

      text-align: center;

      width: 300px;

    }

    input, select, button {

      width: 100%;

      padding: 10px;

      margin: 10px 0;

      border: 1px solid #ccc;

      border-radius: 6px;
```

```
        box-sizing: border-box;
    }
    button{
        background-color: #007bff;
        color: white;
        border: none;
        cursor: pointer;
        font-weight: bold;
    }
    button:hover{
        background-color: #0056b3;
    }
    #result {
        margin-top: 20px;
        font-size: 1.2rem;
        font-weight: bold;
        color: #333;
    }
</style>
</head>
<body>

<div class="calculator-card">
    <h2>JS Calculator</h2>

    <input type="number" id="num1" placeholder="First Number">

    <select id="operator">
        <option value="+">Add (+)</option>
        <option value="-">Subtract (-)</option>
        <option value="*">Multiply (*)</option>
```

```
<option value="/">Divide (/)</option>
```

```
</select>
```

```
<input type="number" id="num2" placeholder="Second Number">
```

```
<button onclick="handleCalculation()">Calculate</button>
```

```
<div id="result">Result: --</div>
```

```
</div>
```

```
<script>
```

```
  // 1. Functional Logic
```

```
  const add = (a, b) => a + b;
```

```
  const subtract = (a, b) => a - b;
```

```
  const multiply = (a, b) => a * b;
```

```
  const divide = (a, b) => b === 0 ? "Cannot divide by zero" : a / b;
```

```
  // 2. The Switch-Case Controller
```

```
  function handleCalculation() {
```

```
    // Get values from the webpage
```

```
    const n1 = parseFloat(document.getElementById('num1').value);
```

```
    const n2 = parseFloat(document.getElementById('num2').value);
```

```
    const op = document.getElementById('operator').value;
```

```
    const display = document.getElementById('result');
```

```
    // Validate inputs
```

```
    if (isNaN(n1) || isNaN(n2)) {
```

```
      display.innerText = "Please enter valid numbers!";
```

```
      return;
```

```
    }
```

```
let result;
```

```
// The Switch-Case you asked for
```

```
switch (op) {  
  case '+': result = add(n1, n2); break;  
  case '-': result = subtract(n1, n2); break;  
  case '*': result = multiply(n1, n2); break;  
  case '/': result = divide(n1, n2); break;  
  default: result = "Invalid Operation";  
}
```

```
// 3. Output the result back to the page
```

```
display.innerText = `Result: ${result}`;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```